Python Developer Guide

Sample training document for the Multi-PDF Technical Documentation Assistant | Version 1.2 | Updated: 10 October 2026

1. Supported Python version

The Acme Automation team supports Python 3.11 and Python 3.12. New projects should use Python 3.12 unless a dependency does not yet support it. Python 3.10 is permitted only for maintaining legacy projects.

2. Virtual environments

Create a project environment with **python -m venv .venv**. On Windows Command Prompt, activate it with **.venv\\Scripts\\activate**. On macOS or Linux, use **source .venv/bin/activate**. Install packages only after activation.

3. Installing packages

Upgrade pip with **python -m pip install --upgrade pip**. Install a package with **python -m pip install package-name**. Record project dependencies with **python -m pip freeze > requirements.txt**. Reinstall them using **python -m pip install -r requirements.txt**.

4. Recommended project structure

• src/ contains application source code.

• tests/ contains automated tests.

• data/ contains local test data; do not commit confidential data.

• requirements.txt records dependencies.

• .gitignore should exclude .venv/, __pycache__/, .env and generated output files.

5. Error handling

Catch specific exceptions whenever possible. Use try/except around operations that may fail, log a useful error message, and avoid silently ignoring exceptions. Do not store API keys directly in Python source code.

6. Configuration and secrets

Store local secrets in environment variables or a local .env file excluded from version control. In production, use the hosting platform's secret manager. Never upload passwords, tokens or private keys to a public repository.

7. Team coding rules

Use meaningful variable names, small functions, docstrings for public functions, and automated tests for important logic. Format code consistently and review changes before merging.